Schelling's Segregation Model: A comprehensive numerical analysis

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Overview

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2. Model Framework

3. Analysis and Results

4. Conclusion and Discussion

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Schelling's Segregation Model

Schelling's Model [Schelling, 1971] ▸ Play Demo (30% tolerance) ▸ Play Demo (70% tolerance)

- Iterative Relocation
- Tolerance Range
- Initial population density / vacancy
- Tipping Point
- **Key Insight**: Mild individual preferences → Extreme collective segregation patterns

Empirical Evaluation of Residential Preferences [Clark, 1991]

- Empirical tolerance curves differ in functional form from Schelling's stylized version
- Consistent with Schelling's conclusion:
  1. Residential preferences still imply that stable integrated equilibria are rare
  2. Neighborhoods exhibit nonlinear transitions and tipping behavior

Our Motivation & Objectives

Motivation:
Prior work lacks a systematic exploration of how these parameters precisely shape the system's long-term dynamics.
Offering implications for the emergence, persistence, and potential mitigation of segregation in real social systems.

Objectives

- Study how inputs (min/max tolerance, population density, grid size) influence outputs (iterations to equilibrium, sameness index)
- Identify critical thresholds (e.g., "tipping point") where small parameter shifts induce large changes in outcomes
- Compare theoretical assumptions with model behavior

Model Framework

Model Mechanism:

- Agent-based Schelling framework on a 2D grid with discrete movements
- Two groups of agents (#, o) measure the fraction of the agents around them that are of the same group $\frac{\#\text{ agents of same type as } i \text{ in neighborhood}}{\#\text{ agents in } i\text{'s neighborhood}}$ and randomly relocate until minimum tolerance level met or max iterations reached

Assumptions:

- Simulation runs on a finite, fixed-size grid with vacancies
- Each agent evaluates the same fixed neighborhood radius (4/8/12)
- Iterations are capped to ensure termination even if the system does not settle
- Movement is local, discrete, and random based on tolerance criteria; without global coordination

Model Framework Cont'd

- **Inputs**:
  1. Minimum & Maximum tolerance levels for each group
  2. Population density for each group: $\text{density}_i, i \in \{\#, o\}$
  3. Grid size
  4. Neighborhood size (radius)
- **Outputs**:
  1. Number of Iterations
  2. Sameness index: $\dfrac{\sum_{\text{agent } i} \frac{\#\text{ agents of same type as } i \text{ in neighborhood}}{\#\text{ agents in } i\text{'s neighborhood}}}{\#\text{ agents in the grid}}$

Our Initial Assumptions

1. If the sum of minimum tolerances across groups **exceeds 1**, the system will fail to converge (i.e. iteration goes to ∞).
2. Setting equal minimum tolerance for both groups is the representative case.

These assumptions will be tested against the model's results!

Numerical Analysis - Parameter Space

Invariants:

1. Grid size: **50**
2. Neighborhood size: **12**

Parameters:



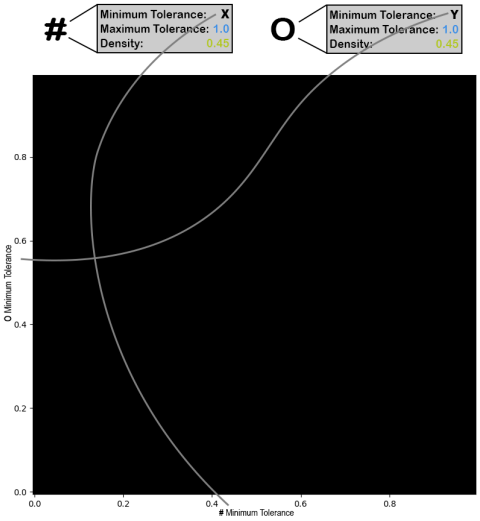
6 different input parameters

Far too large an input space to explore for this presentation!

Numerical Analysis - Our Cross-Section

To narrow down our parameter space, we will vary only two of the parameters:

1. The **minimum tolerances** of **#** and **O** agents (**X** and **Y**) will vary independently
2. Both groups of agents will have their **maximum tolerances** set to **1.0** (i.e. no effective maximum tolerance)
3. Both groups of agents will have **densities** of **0.45** (i.e. equal amounts with 10% vacancies)



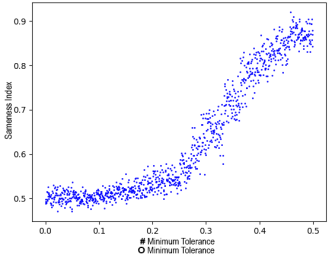
Numerical Analysis - Emerging Complexity

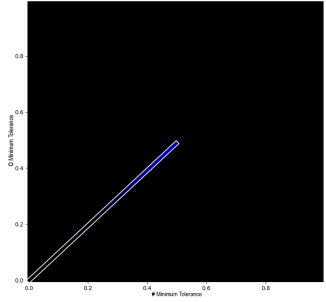
Our Initial Assumptions

1. If the sum of minimum tolerances across groups **exceeds 1**, the system will fail to converge (i.e. iteration goes to ∞).
2. Setting equal minimum tolerance for both groups is the representative case.

Simple measurement with identical similarity scores produces sensible results

However, this is only a **small part** of our cross-section; expanding analysis to the whole section yields a more interesting result...





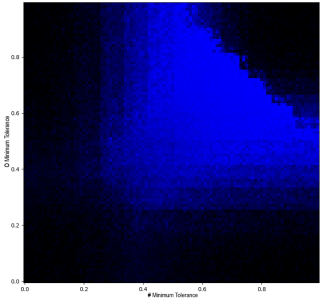
Numerical Analysis - Emerging Complexity (Cont'd)

Our Initial Assumptions

1. If the sum of minimum tolerances across groups **exceeds 1**, the system will fail to converge (i.e. iteration goes to ∞).
2. Setting equal minimum tolerance for both groups is the representative case.

Varying the minimum tolerances of the two populations **independently** yields a more complex pattern

Question: What is the cause of the triangular region in the upper-right (corresponding to extremely high minimum tolerances for both populations) which has *low* similarity index?



Numerical Analysis - Emerging Complexity (Cont'd)

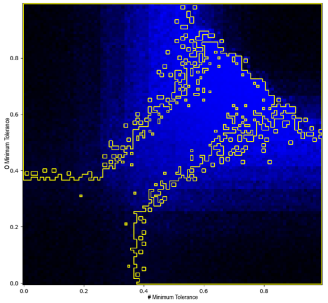
Our Initial Assumptions

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Varying the minimum tolerances of the two populations **independently** yields a more complex pattern

Yellow shows the boundary between trials which did and did not terminate (converge).

- The dark upper-right region consists of trials which did not terminate
- However, some other trials which did not terminate (in the corners) have high similarity indices - indicating different behavior!



Numerical Analysis - Iteration Analysis

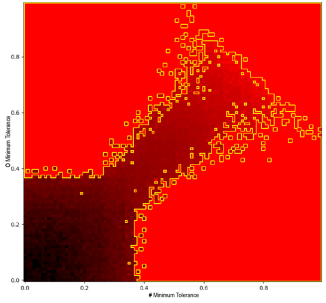
Our Initial Assumptions

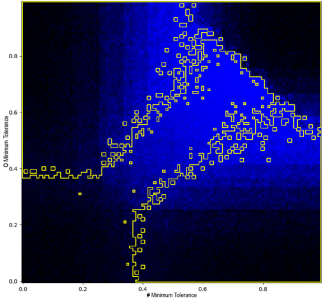
1. ~~If the sum of minimum tolerances across groups **exceeds 1**, the system will fail to converge (i.e. iteration goes to ∞).~~
2. ~~Setting equal minimum tolerance for both groups is the representative case.~~

Graphing the **log iteration count** helps to illuminate this behavior.

- This is the number of moves that were required before the system terminated (capped at 20,000)

Both of our initial assumptions are proven false by this result!



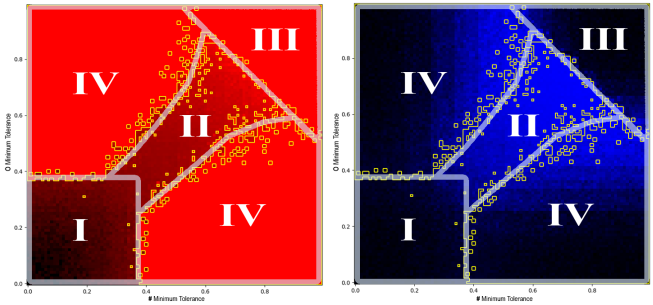


Numerical Analysis - Qualitative Regions

Through simple visual analysis, we can identify **four qualitatively distinct regions** on this graph:

- **I:** The bottom-left square
- **II:** The diagonal column
- **III:** The upper-right triangle
- **IV:** The corner regions

With our remaining time, we will quickly summarize a week's worth of intensive research to **characterize the distinct behavior** of each of these regions.



Numerical Analysis - Boundary Analysis

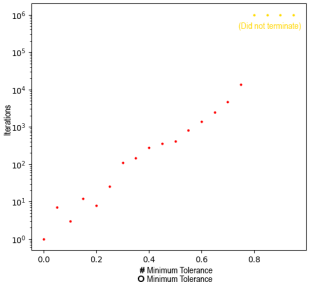
The boundary between regions **II** and **III** occurs along the diagonal at **X=Y=0.75** exactly.

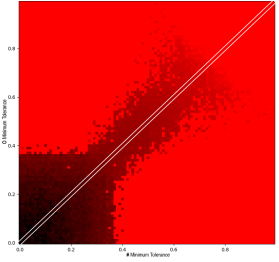
Since the boundary has a **slope of -1** on the graph, this means the cutoff is **X+Y=1.5**

- This is contrasted with our initial assumption, which was that this boundary would exist at X+Y=1.0

We additionally observe that iteration counts along the diagonal follow an **exponential** (log-linear) pattern up to the tipping point!

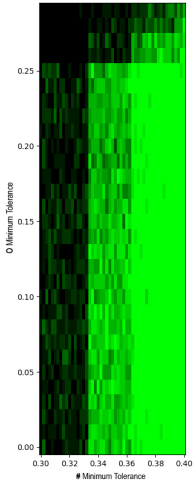
- The maximum number of iterations observed in a convergent trial is **20,000**, so this will be used as our soft "cap" to save on model runtime





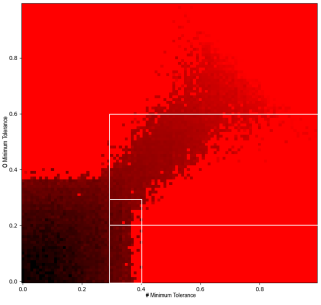
Numerical Analysis - Boundary Analysis (Cont'd)

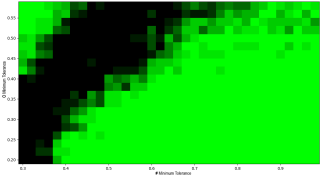
Depicted below in **green** are the shares of **10 trials** at each set of coordinates which **did not converge**.



- The boundary between regions **I** and **IV** occurs in stages at **max(X,Y)=0.333** and **max(X,Y)=0.365**
- The boundary between regions **II** and **IV** is fuzzy and difficult to precisely characterize

  (Mathematically characterizing the **II-IV** boundary is an **open question**!)

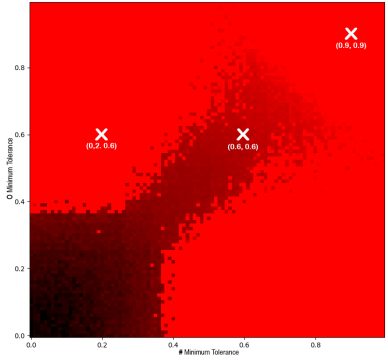




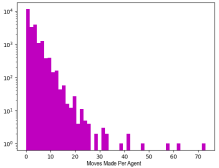
Numerical Analysis - Behavior Analysis - Convergence

Plotting the **histogram** of **movement counts** for each agent reveals the reason for the difference in behavior:

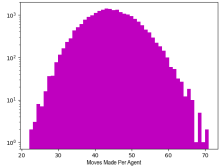
- Movements which **converge** follow a characteristic **power law**
- Movements which **do not converge** follow a characteristic **normal distribution**



(0.6,0.6)



(0.9,0.9)

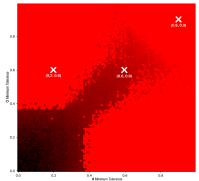


Numerical Analysis - Behavior Analysis - Convergence (Cont'd)

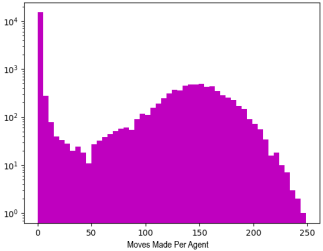
Plotting for parameters in Region **IV** gives a distribution with **both** log-linearly- and normally-distributed components

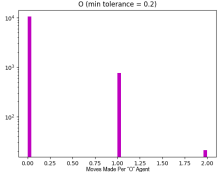
Decomposing the histogram into the **lower** (left) and **higher** (right) minimum tolerance agent types:

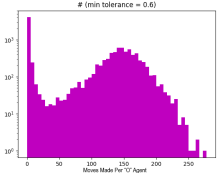
- The lower minimum-tolerance agents display **convergent** behavior
- The higher minimum-tolerance agents are divided, with some displaying convergent behavior, and others **never settling**



(0.2,0.6)





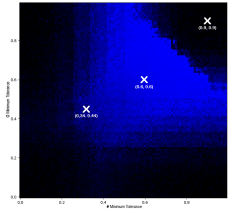


Numerical Analysis - Behavior Analysis - Similarity

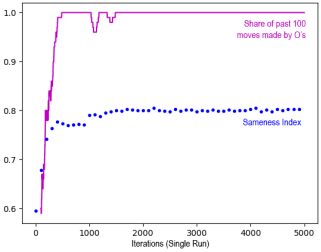
Why do similarity indices reach high values in Region **IV**, but not in Region **III**?

We plot the change in **similarity index** and **type-ratio of moves** over time for a single trial:

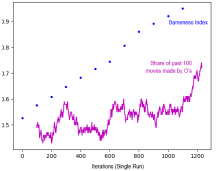
- Region **IV**: Similarity index *increases* while *both* agent types move, and *does not substantively change* when *only one* agent type is still moving
- Region **III**: Similarity index never increases in the first place, movements appear undirected



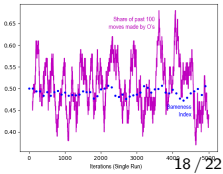
(0.34,0.44)



(0.6,0.6)



(0.9,0.9)



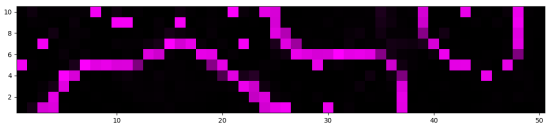
Numerical Analysis - Behavior Analysis - Similarity (Cont'd)

Why does the similarity index **increase and then stop changing** in Region **IV**, and **never meaningfully increase** in Region **III**?

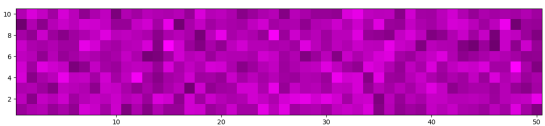
Plotting a **movement heatmap** reveals that:

- Non-transient movement is confined to **small areas** in a Region **IV** trial
- The **whole area** continues to shuffle indefinitely in a Region **III** trial

(0.5,0.7)



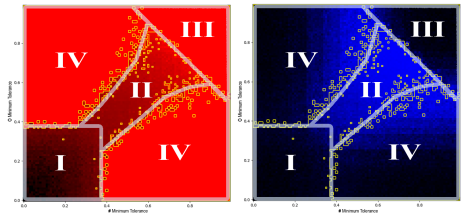
(0.9,0.9)



Numerical Analysis - Summary

We can therefore fully characterize the behavior of the four distinct regions of our cross-section of the parameter space:

- **Region I** is a square gradated-ly bounded between **max(X,Y) = 0.333** and **0.365**, and displays **fast convergence** & **low similarity**
- **Region II** is the region of convergence between Region I and **X+Y = 1.5** which is not obstructed by Region IV; it displays **exponentially increasing** convergence along the diagonal (up to $\sim$ 20,000 iterations) and **high similarity**
- **Region III** is the region above **X+Y = 1.5**, and displays **random-noise movement** with **non-convergence** for both agent types, and **low similarity**



- **Region IV** is characterized by **unequal minimum tolerances**; it displays **non-convergence** in one agent type and **convergence** in the other, and **high similarity**

Conclusion and Discussion

- We closely explored how parameters in the Schelling model jointly shape long-run dynamics using numerical simulations
- We studied a 2-dimensional cross-section of the full 6-dimensional input space
- This cross-section divides into four distinct dynamical regions, characterized by **sameness index**, **iteration counts**, and **convergence** vs. **non-convergence**
- The model's behavior departs from initial assumptions, motivating a more general theoretical interpretation
- We identified numerical tipping points separating qualitatively different regimes

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